Strategy-proofness versus collusion-proofness

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The design tradeoff

- Market designers face a tradeoff in designing mechanisms
  1. Efficiency: allocating resources to those who value them most
  2. Budget Balance: running no deficit
  3. Incentive Compatibility: preventing manipulation
  4. Individual Rationality: ensuring voluntary participation
- Most classic literature focuses on *individual deviation*.
- Collusion: bidding rings and cartels coordinate to manipulate mechanisms $\rightarrow$ loss of efficiency and revenue

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Loss for the public

"What really matters in auction design are the same issues that any industry would recognize as key concerns: ... collusive behavior..."

—Klemperer (2002)

- *US highway construction* (Porter and Zona, 1993)
  $\rightarrow$ Winning bids inflated by cartel members using fake bids.
- *Ohio school milk* (Porter and Zona, 1999)
  $\rightarrow$ Schools overpaid by 6.5% due to territorial allocation.
- *Japanese public works* (Kawai and Nakabayashi, 2022)
  $\rightarrow$ Collusion affected \$8.6 billion in contracts (19% projects).

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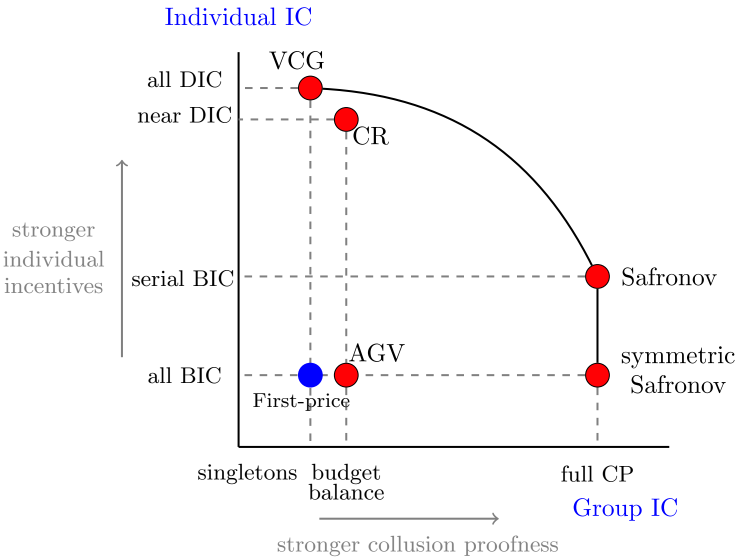
Efficient mechanism design

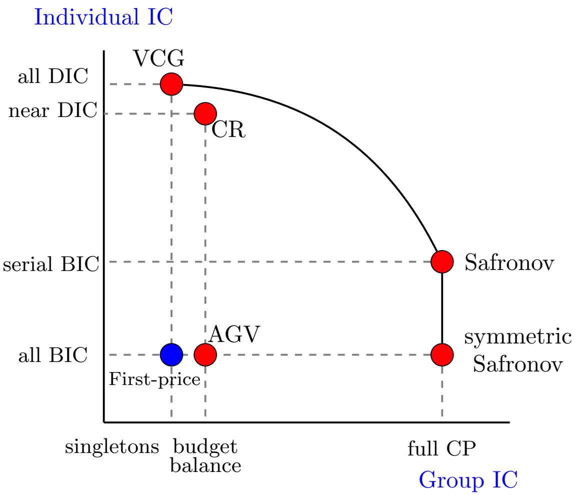
The benchmarks

Existing 6 *efficient* mechanisms under IPV sit at extremes



Position: Bottom-Left ($\equiv 0$)

- works only in auction environment with symmetric agents.
- has n BIC.
- fails ex post budget balance.
- is collusion-proof only against *singletons*.



1. First-price auction Vickrey (1961); Riley and Samuelson (1981)

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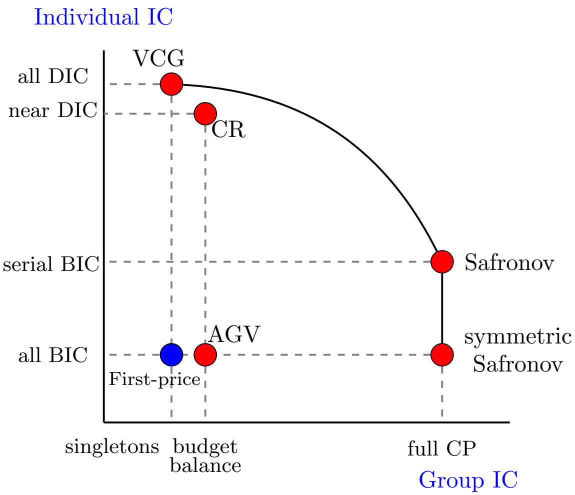

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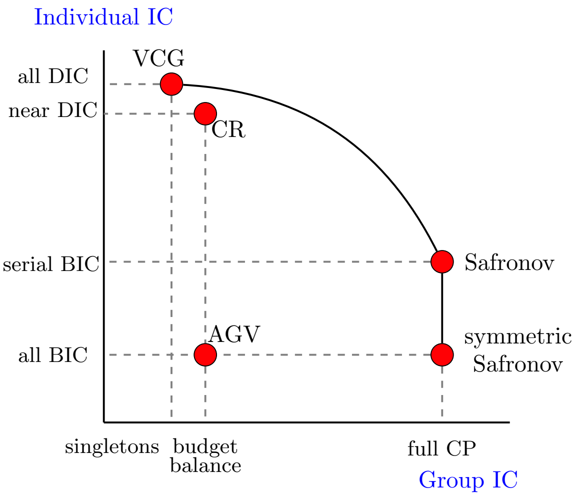

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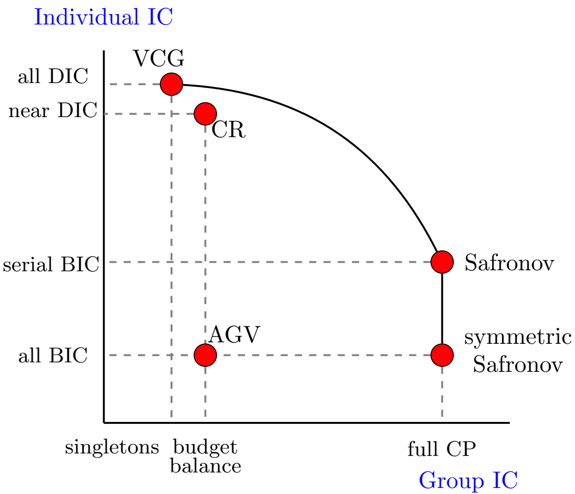

3. AGV mechanism Arrow (1979); d'Aspremont and Gérard-Varet (1979)

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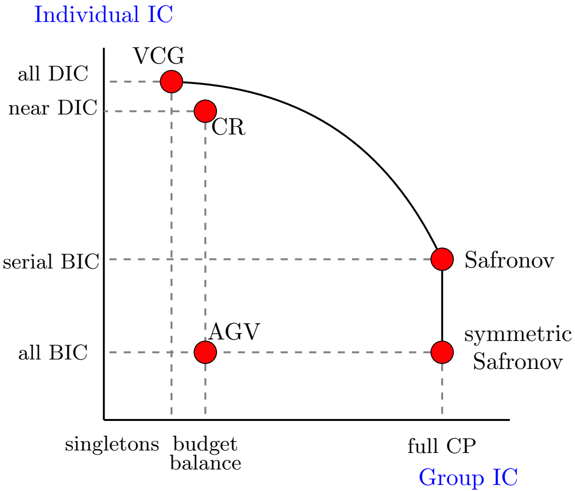
- CP against *Grand Coalition*, and singletons.



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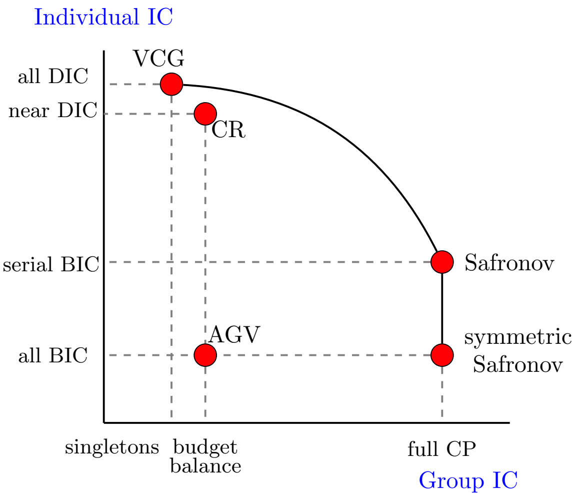

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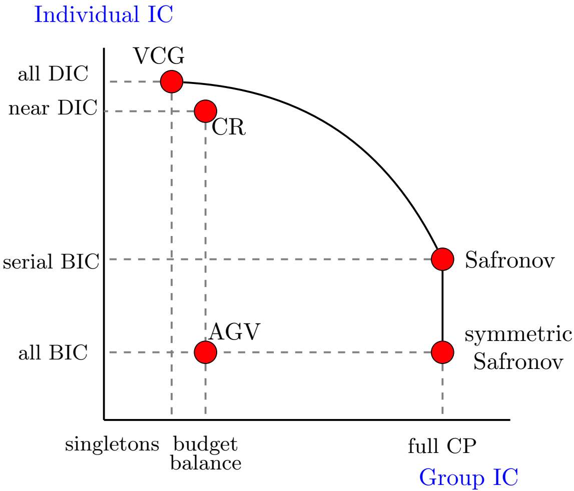

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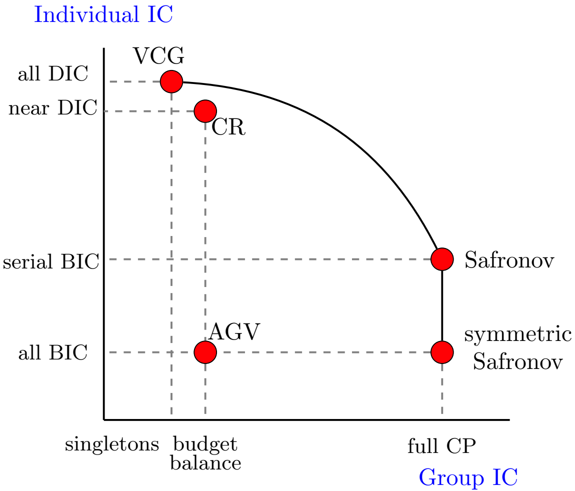
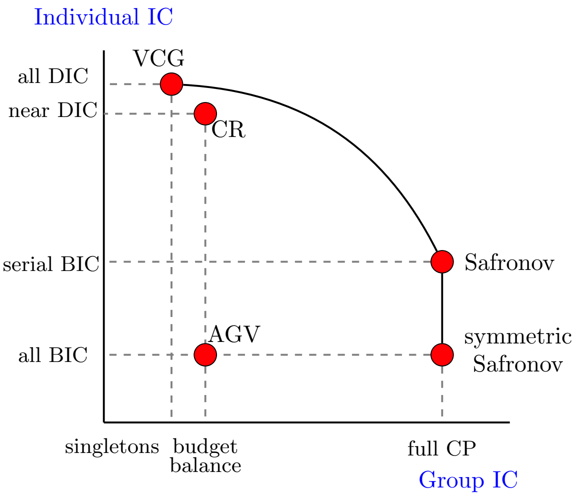


5. Safronov mechanism Safronov (2018)

Position: near Bottom-Right.

- has Serial BIC (including 1 DIC & 1 BIC).
- balances budget ex post.
- has full CP, i.e. proof against all coalitions.



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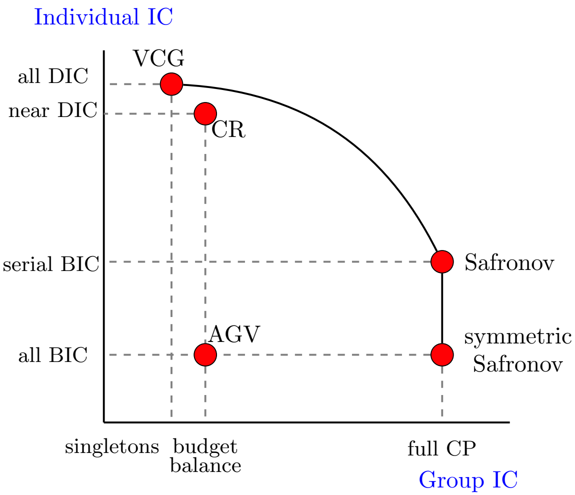

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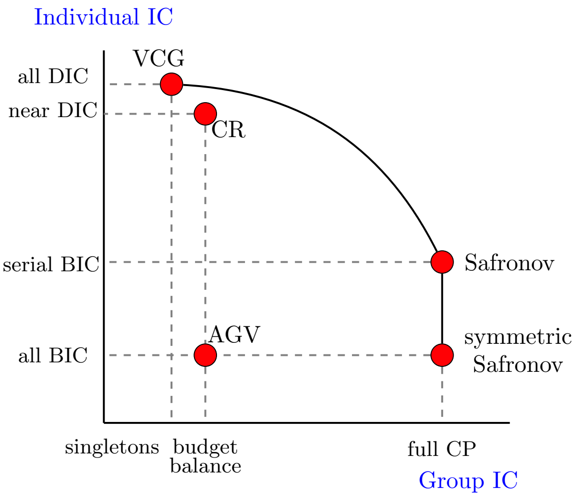

6. Symmetric Safronov mechanism Safronov (2018)

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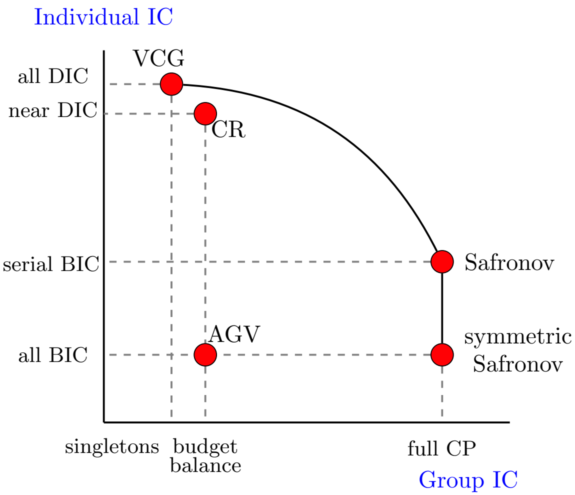


What have we done so far?

Questions

Question 1. The frontier

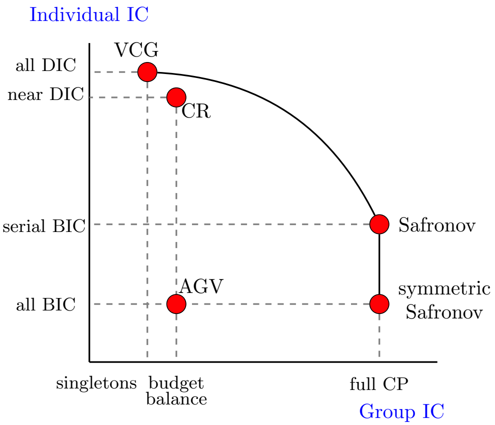
Is there a frontier displaying the tradeoff between SP & CP?

Question 2. New designs

Can we design mechanisms that dominate existing benchmarks?

Question 3. Unification

Are AGV, CR, Safronov derived from a single principle?



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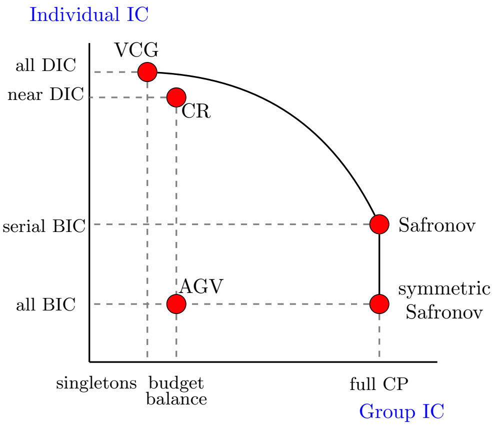

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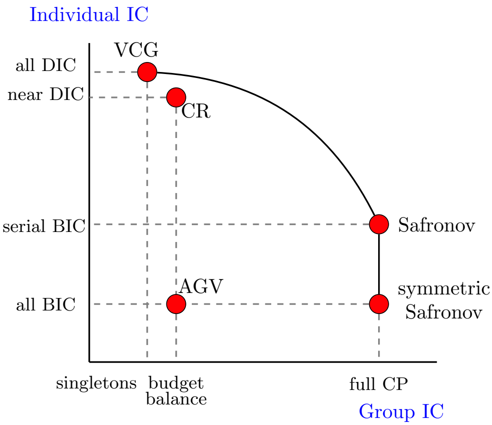

Preview of contributions

1. New definitions: *more/less collusion proof* (by conference structure); *more/less strategy proof* (by MIC)

2. Tradeoff:
   - establish frontier between SP and CP for efficient mechanisms
   - propose new designs: α, β, and sequence of designs β_k

3. Construction: all well-known mechanisms are projections of VCG onto subspaces of incentive constraints.
   - Impossibility: a wishlist of incentive constraints are incompatible

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Setup & definitions

Environment

Primitives

- Agents $N = \{1, \dots, n\}$
- Finite set of alternatives $\mathcal{X}$
- Independent private types $\theta_i \in \Theta_i$ drawn from $\pi_i(\theta_i)$
- Quasilinear utility $u_i(x, \theta_i) + t_i$ with $x \in \mathcal{X}$ and $t_i \in \mathbb{R}$

Ex post efficiency

- Efficient allocation rule χ selects x to maximize social welfare[1]

$$\chi(\boldsymbol{\theta}) \in \arg\max_{x \in \mathcal{X}} \left\{ \sum_{i \in N} u_i(x, \theta_i) - h(x) \right\} \quad \rightsquigarrow \text{Welfare func. } W(\boldsymbol{\theta}).$$

- An efficient mechanism is a pair $(\chi, \mathbf{T})$, where $T_i(\boldsymbol{\theta})$ specifies the transfer to agent i given report $\boldsymbol{\theta}$.
- Ex post payoff from reporting $\hat{\theta}_i$ given true type θ_i and opponents' reported types $\boldsymbol{\theta}_{-i}$

$$V_i(\hat{\theta}_i, \theta_i; \boldsymbol{\theta}_{-i}) = u_i\big(\chi(\hat{\theta}_i, \boldsymbol{\theta}_{-i}), \theta_i\big) + T_i(\hat{\theta}_i, \boldsymbol{\theta}_{-i}).$$

[1] Assumes $h(x) = 0$, yet covers revenue-optimal auctions.

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Strategy proofness

Standard definitions

- *DIC*: Truthtelling is optimal regardless of others' strategies.
- *BIC*: Truthtelling is optimal in expectation over others' types.

Definition: Mid Incentive Compatibility (MIC)

Agent i is MIC w.r.t. subset S if truth-telling is a best response for i, *conditional* on agents in S being truthful; that is, for all θ_i, regardless of $\boldsymbol{\theta}_{-S\cup i}$

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Special cases:

- DIC $\equiv$ MIC when $|S| = 0$
- BIC $\equiv$ MIC when $|S| = n-1$
- almost DIC $\equiv$ MIC when $|S| = 1$

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Collusion proofness

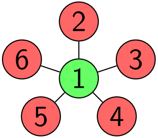
Definition: Conference Structure & Conference-Proofness

A conference structure is a set $\mathcal{C}$ of feasible collusive coalitions.

A mechanism is collusion-proof against $\mathcal{C}$ if no coalition $C \in \mathcal{C}$ can profit from a joint misreport. That is, for every $C \in \mathcal{C}$ and for every realized $\boldsymbol{\theta}_C \in \Theta_C$

$$\boldsymbol{\theta}_C \in \arg \max_{\hat{\boldsymbol{\theta}}_C \in \Theta_C} \int_{\Theta_{-C}} \sum_{i \in C} V_i(\hat{\theta}_i, \theta_i; \hat{\boldsymbol{\theta}}_{C \setminus i}, \theta_{-C}) \, d\pi_{-C}(\theta_{-C}).$$

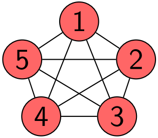
Star Network
(pivotal agent 1)



Collusion requires the Hub.

$\mathcal{C} = \{\{i\} : i \text{ from } 1 \text{ to } 6\} \cup \{C : 1 \in C\}$

Complete Network
(full collusion)

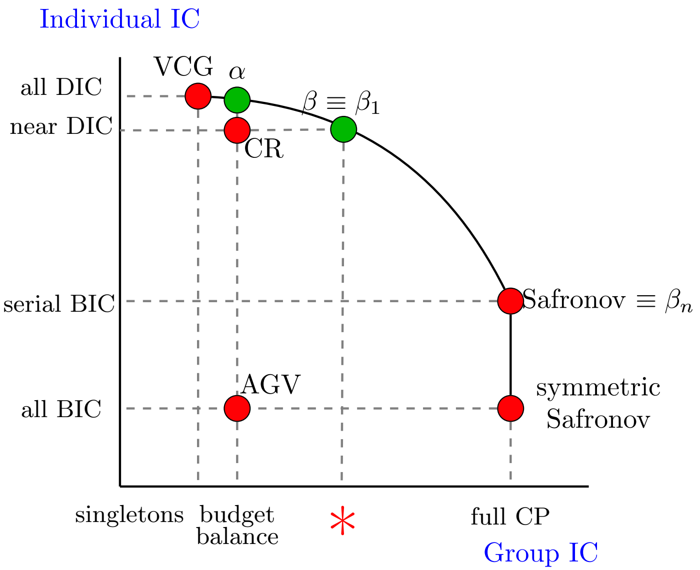


Any subset can collude.

$\mathcal{C} = 2^N$

Improving CR mechanisms

Incentive frontier



Groves schemes

- Groves payments align individual incentive with $W(\boldsymbol{\theta})$: n DIC.
- Agent i gets

$$T_i^G(\boldsymbol{\theta}) = \underbrace{W_{-i}(\boldsymbol{\theta})}_{W(\boldsymbol{\theta})-u_i(\chi(\boldsymbol{\theta}),\theta_i)} - \underbrace{h_i(\boldsymbol{\theta}_{-i})}_{\text{chosen freely}}$$

  - VCG: pay externality, $h_i(\boldsymbol{\theta}_{-i}) = \inf_{\theta_i} W(\theta_i, \boldsymbol{\theta}_{-i})$: n ex post IR
  - AGV: pay average externality: n BIC, budget balanced

Crémer-Riordan mechanism

- also known as Stackelberg mechanism
- *Incentives*:
  - Agents $2, \ldots, n$: Groves transfers ($(n-1)$ DIC)

$$\begin{aligned} T_2^{\mathrm{CR}}(\boldsymbol{\theta}) &= W_{-2}(\boldsymbol{\theta}) - W(\theta_1), \\ T_3^{\mathrm{CR}}(\boldsymbol{\theta}) &= W_{-3}(\boldsymbol{\theta}) - W(\theta_1), \\ &\cdots \\ T_n^{\mathrm{CR}}(\boldsymbol{\theta}) &= W_{-n}(\boldsymbol{\theta}) - W(\theta_1). \end{aligned}$$

  - Agent 1: Residual Claimant (1 BIC)

$$T_1^{\mathrm{CR}}(\boldsymbol{\theta}) = W_{-1}(\boldsymbol{\theta}) + (n-1)W(\theta_1) - (n-1)W(\boldsymbol{\theta}).$$

- *Collusion*: vulnerable. no protection against coalitions.
- *Budget*: ex post balanced.

Can we do better?

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The α-Mechanism

- Target: Improve the incentives of the residual claimant
- α-mechanism: ▸ See Appendix
  - Agents $3, \ldots, n$: Standard Groves (DIC)
  - Agent 1: Residual Claimant (Groves, DIC)
  - Agent 2: Residual Claimant (aDIC)
- α-mechanism $\approx$ VCG mechanism

Theorem 1: α-mechanism

The α-mechanism achieves EBB, $(n-1)$ DIC $+$ 1 aDIC.

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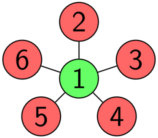
The β-Mechanism

- Target: Prevent "Hub-and-Spoke" collusion common in procurement settings
- β-mechanism: See Appendix
  - Agents $2, \ldots, n$ (Periphery): Groves, Utility Guarantee (DIC).
  - Agent 1 (Hub): Residual claimant (BIC).

Theorem 2: Star CP and β-mechanism

The β-mechanism is EBB, $(n-1)$ DIC $+$ 1 BIC, and collusion proof against $\mathcal{C}$ generated by a star network.



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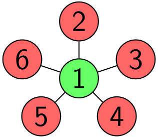
Theorem 2: Star CP and β-mechanism

The β-mechanism is EBB, $(n-1)$ DIC $+$ 1 BIC, and collusion proof against $\mathcal{C}$ generated by a star network.

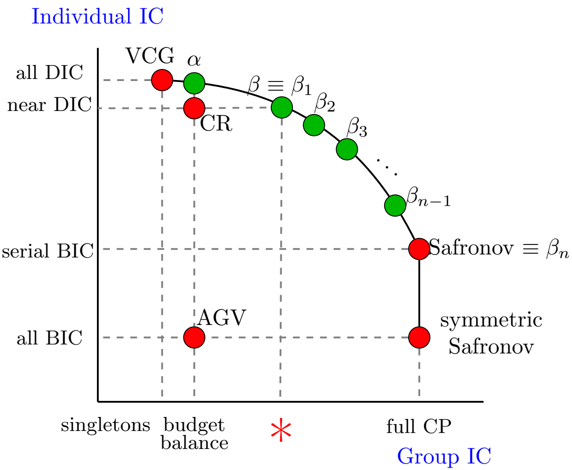


The incentive frontier

Interpolation: The β_k Sequence

- What if the network is denser than a star?
- We construct a sequence $\{\beta_2, \beta_3, \ldots, \beta_n\}$.
- β_n (Safronov) is fully collusion proof

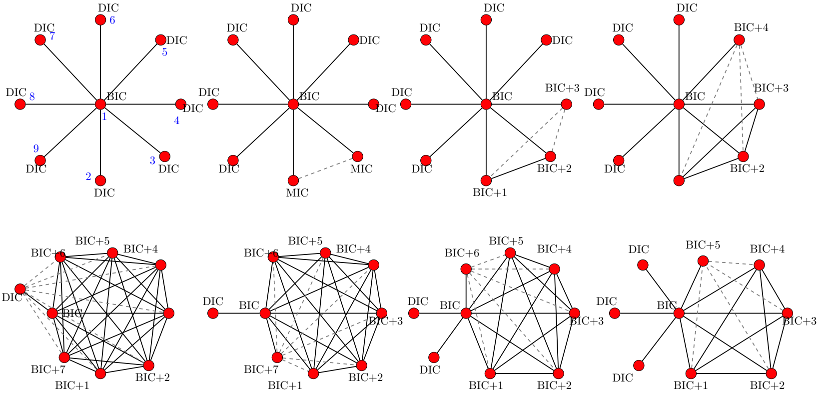
See Appendix



- As feasible coalitions grow, we must weaken individual incentives to satisfy collusion proofness.

See Appendix

$n = 9$ agents



Interim IR

Definition: Interim Individual Rationality (IIR)

A mechanism is IIR if for every agent i and every type θ_i

$$V_i(\theta_i) = \mathbb{E}_{\boldsymbol{\theta}_{-i}}[u_i(\chi(\boldsymbol{\theta}), \theta_i) + T_i(\boldsymbol{\theta})] \geq R_i(\theta_i)$$

Theorem 3

If there exists a BIC, expected BB, efficient mechanism that is IIR $\Longrightarrow$ there exists a mechanism that is Serial IC, Ex Post BB, Efficient, IIR + Collusion Proof

CGK (1987): With roughly equal shares ($r_i \approx 1/n$), there is an efficient dissolution mechanism that satisfies:

1. Ex Post Budget Balance
2. Interim Individually Rationality
3. Serial BIC
4. Collusion Proof against 2^N

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How to construct mechanisms?

The projection principle

- We can view efficient mechanisms as points in a Hilbert space of transfer rules.
- Inner product

$$\langle T^A, T^B \rangle = \sum_{i \in N} \int_{\Theta} T_i^A(\theta) T_i^B(\theta) \, d\pi(\theta)$$

- Incentive constraints (e.g, EBB, CP against C, MIC) form affine subspaces. However, CP and MIC are linear inequalities.
- MIC, CP become linear constraints by Envelope theorem under smoothly-connected type space (Holmström, 1979)

Theorem 4: Mechanisms as Projections

AGV, Safronov, α and the β-family are all orthogonal projections of Groves schemes onto the intersection of incentive constraints.

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Conclusion

Summary

- Frontier established: We charted the trade-off between Individual IC and Group IC.
- Dominated benchmarks: The α and β mechanisms dominate the classic CR design in either dimensions.
- Unification: Projection unifies classic mechanisms and generates new designs.
- Much work remains: precise ordering of conference structure in collusion, other mechanisms inside and on the frontier etc.

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Appendix: α-mechanism

$$
\begin{aligned}
T_1^\alpha(\boldsymbol{\theta}) &= W_{-1}(\boldsymbol{\theta}) - W_{-1} - (n-1)W(\boldsymbol{\theta}_{-1}) + (n-1)W, \\
T_2^\alpha(\boldsymbol{\theta}) &= W_{-2}(\boldsymbol{\theta}) - W_{-2} + (n-1)W(\boldsymbol{\theta}_{-1}) - (n-1)W(\boldsymbol{\theta}), \\
T_3^\alpha(\boldsymbol{\theta}) &= W_{-3}(\boldsymbol{\theta}) - W_{-3}, \\
\cdots & \\
T_n^\alpha(\boldsymbol{\theta}) &= W_{-n}(\boldsymbol{\theta}) - W_{-n}.
\end{aligned}
$$

Return

Appendix: β-mechanism

$$
\begin{aligned}
T_1^{\beta}(\boldsymbol{\theta}) &= W_{-1}(\boldsymbol{\theta}) - (n-1)W(\boldsymbol{\theta}) + \sum_{j=2}^{n} W(\boldsymbol{\theta}_{-j}) - W + u_1 \\
T_2^{\beta}(\boldsymbol{\theta}) &= W_{-2}(\boldsymbol{\theta}) - W(\boldsymbol{\theta}_{-2}) + u_2 \\
T_3^{\beta}(\boldsymbol{\theta}) &= W_{-3}(\boldsymbol{\theta}) - W(\boldsymbol{\theta}_{-3}) + u_3 \\
&\cdots \\
T_n^{\beta}(\boldsymbol{\theta}) &= W_{-n}(\boldsymbol{\theta}) - W(\boldsymbol{\theta}_{-n}) + u_n.
\end{aligned}
$$

Return

Appendix: Safronov mechanism

Safronov (2018) proposes the following mechanism $(\chi, \mathbf{T}^S)$

- Ordering:

$$\underbrace{n}_{\text{DIC}} \rightarrow \underbrace{n-1}_{\text{MIC: } S=\{n\}} \rightarrow \underbrace{n-2}_{\text{MIC: } S=\{n,n-1\}} \rightarrow \ldots \rightarrow \underbrace{2}_{\text{MIC: } S=\{3:n\}} \rightarrow \underbrace{1}_{\text{BIC}}$$

- Transfer rules are based on staircase architecture[2]:

$$T_1^S(\boldsymbol{\theta}) + u_1(\chi(\boldsymbol{\theta}), \theta_1) - u_1 = W(\theta_1) - W,$$
$$T_2^S(\boldsymbol{\theta}) + u_2(\chi(\boldsymbol{\theta}), \theta_2) - u_2 = W(\theta_1, \theta_2) - W(\theta_1),$$
$$\vdots$$
$$T_n^S(\boldsymbol{\theta}) + u_n(\chi(\boldsymbol{\theta}), \theta_n) - u_n = W(\theta_1, \ldots, \theta_n) - W(\theta_1, \ldots, \theta_{n-1}).$$

- Utility Guarantee + Budget Balance $\rightarrow$ CP against 2^N Return

[2]The initial construction in Safronov (2018) is based on bilateral externalities. We propose this simpler construction amenable to generalization.

Appendix: β_k mechanism

$$
\begin{aligned}
T_1^{\beta_k}(\boldsymbol{\theta}) + u_1(\boldsymbol{\theta}) - u_1 &= W(\boldsymbol{\theta}) - (n-k+1)W(\boldsymbol{\theta}) \\
&\quad + \sum_{j=2}^{n-k} W(\boldsymbol{\theta}_{-j}) - W(\boldsymbol{\theta}_{n-k+1:n}) \\
T_2^{\beta_k}(\boldsymbol{\theta}) + u_2(\boldsymbol{\theta}) - u_2 &= W(\boldsymbol{\theta}) - W(\boldsymbol{\theta}_{-2}) \\
&\cdots \\
T_{n-k}^{\beta_k}(\boldsymbol{\theta}) + u_{n-k}(\boldsymbol{\theta}) - u_{n-k} &= W(\boldsymbol{\theta}) - W(\boldsymbol{\theta}_{-(n-k)}) \\
T_{n-k+1}^{\beta_k}(\boldsymbol{\theta}) + u_{n-k+1}(\boldsymbol{\theta}) - u_{n-k+1} &= W(\boldsymbol{\theta}_{n-k+1:n}) - W(\boldsymbol{\theta}_{n-k+2:n}) \\
&\cdots \\
T_{n-1}^{\beta_k}(\boldsymbol{\theta}) + u_{n-1}(\boldsymbol{\theta}) - u_{n-1} &= W(\boldsymbol{\theta}_{n-1:n}) - W(\theta_n) \\
T_n^{\beta_k}(\boldsymbol{\theta}) + u_n(\boldsymbol{\theta}) - u_n &= W(\theta_n) - W.
\end{aligned}
$$

Return